# LESSON 5: FLOWER, FRUITS AND SEEDS

The flower's whorls, how its parts become fruits and seeds, and their types



### What is a flower for?

The **flower** is the plant's **reproductive part**. Its biological function is to **facilitate reproduction**, usually by providing a mechanism for the **union of sperm with eggs**. After fertilization the **ovary becomes the fruit** and the **ovules become the seeds**.

- **Smallest flower** – **Wolffia** | **Largest flower** – **Rafflesia**

## THE FLOWER AND ITS FOUR WHORLS

- Four parts, outside → in: **calyx** → **corolla** → **androecium** → **gynoecium**
- **Non-essential** – **calyx, corolla** | **Essential** – **androecium, gynoecium** (the reproductive whorls)

### CALYX (1ST, OUTERMOST WHORL)

- Unit: **sepal**; usually **green**; non-essential
- **Petaloid** – calyx is **brightly colored**, like petals
- **Polysepalous** – **free** sepals; **rose**
- **Gamosepalous** – **fused** sepals; **Datura**

> **Note:** The slide titled "Polysepalous" says "petals are free"; it means **sepals**.

### COROLLA (2ND WHORL)

- Unit: **petal**; **variously colored**; non-essential
- **Polypetalous** – **free** petals; **rose**
- **Gamopetalous** – **fused** petals; **Datura**
- **Apetalous** – petals **absent** (rare); **Brassica rapa**

> **Note: Poly** = free (separate), **gamo** = fused, **a-** = without. **-sepalous** = calyx, **-petalous** = corolla.

## ANDROECIUM AND GYNOECIUM

### ANDROECIUM (3RD WHORL, MALE)

- Unit: **stamen**; essential; **male** reproductive part
- **Anther** – head of the stamen; holds the **pollen grains** (for pollination)
- **Filament** – the **stalk** that holds the anther

### GYNOECIUM (INNERMOST WHORL, FEMALE)

- Unit: **carpel**; essential; **female**; after fertilization it develops into the **fruit**
- **Stigma** – top of the carpel, **sticky**; pollen sticks to it
- **Style** – the **tube** the sperm cell travels down to reach the ovary
- **Ovary** – where **fertilization** happens; produces the eggs; has the **ovary wall** and **ovules**
  - **Ovary → fruit** | **Ovary wall → outer coat of the fruit** | **Ovules → seeds**

### Can-do / can't-do clues

Pollen won't stick → **stigma**. Pollen sticks but sperm never reaches the ovary → **style**. No pollen → **anther**. Anther not held up → **filament**. Fruit but no seeds → **ovules**. Seeds but no fruit coat → **ovary wall**.

## TYPES OF FLOWERS

- **Complete** – **all 4 whorls**; most **bisexual** flowers are complete
- **Incomplete** – **any whorl missing**; **all unisexual flowers are incomplete**
- **Bisexual** – **androecium + gynoecium** in the same flower
- **Unisexual**: **male = staminate** (androecium only) | **female = pistillate** (gynoecium only)

## FRUITS

- Fruits are found **only in angiosperms**
- **True fruit** – from the **ovary only** | **Accessory fruit** – ovary **+ receptacle, petals, sepals** (**pineapple**)
- **Simple** – from a **single ovary** (one pistil), one or more carpels; **fleshy or dry**; true or accessory
- **Aggregate** – **many pistils of the same flower**; units may be berries
- **Multiple** – pistils from **many flowers**, often with accessory parts (**mulberry**)

## FLESHY FRUITS (WALL SOFT AT MATURITY)

- **Drupe** – **one seed**, from a superior ovary; **exocarp** thin skin, **mesocarp** fleshy or fibrous, **endocarp hard and stony** (cherry, peach, plum)
  - Skin = outer ovary layer, flesh = middle, pit = **inner** layer
  - **Coconut** – drupe with **fibrous** walls; liquid and solid **endosperm**
- **Berry** – ovary wall (at least its inner part) **enlarged and juicy** (grape, banana, gooseberry, tomato)
  - **Hesperidium** – **leathery rind**; **septa = number of carpels** (orange, lemon)
  - **Pepo** – **hard rind** (watermelon, gourds, squash, pumpkin, cucumber)
- **Pome** – French **pomme**, apple; **accessory**: carpels surrounded by the floral tube/receptacle; **5+ carpels**; **core from the ovary**, **pulp from the perianth** (apple, pear, mountain ash)

## DRY FRUITS (COAT DRY AT MATURITY)

- **Dehiscent** – **open** along definite seams to shed seeds | **Indehiscent** – **do not open**; many are **one-seeded**

### DEHISCENT

- **Legume** – **1 carpel**, splits along **both** dorsal and ventral **sutures**; two rows of ovules (pea, bean)
  - **Peanut** – one of the few legumes that **does not split**; probably because it **develops in the soil**
- **Follicle** – **1 carpel**, splits along **one suture** (larkspur, columbine, milkweed, bay magnolia)
- **Capsule** – **several carpels**; lily splits into sections = number of carpels
  - **Loculicidal** – along the **outer median line** (lily, Viola, Yucca, Datura)
  - **Poricidal** – opens by **round holes** (poppy, Papaver)
  - **Denticidal** – **small teeth** at the top (corn cockle, Agrostemma githago)
  - **Septicidal** – along the **septa**, opens at the top (Dutchman's pipe, Aristolochia)
  - **Silique** – **long, slender**, 2 carpels (mustard family, Brassicaceae/Cruciferae) | **Silicle** – **short, broad**, 2 carpels
  - **Pyxis** – **circumscissile**: the top comes off like a lid (plantain)

### INDEHISCENT

- **Achene** – one seed, attached to the fruit wall at **one point**; seed free of the **pericarp** (buttercup, dandelion, sunflower)
- **Nut** – like an achene, wall **greatly thickened and hard** (beech, chestnut, oak, hazel); acorn cup = **involucre**; walnut and hickory sometimes called drupes (husk)
- **Samara** – **winged** achene; one or two seeds; **wind dispersal** (elm, maple, ash)
- **Grain (caryopsis)** – **seed coat fused to the pericarp** (wheat, corn)
- **Schizocarp** – from a compound pistil; splits into **mericarps** that look like small fruits

> **Note: Legume vs follicle** = two seams vs one. **Achene vs grain** = seed free (one point) vs fused. **Achene vs nut** = thin vs very hard wall.

## SEEDS

- **Seed** – a **fertilized matured ovule** with a **seed coat**; the **propagating material**; an **embryonic plant** in a protective covering; completes reproduction in seed plants
- Three parts: **embryo**, **endosperm** (stored food), **seed coat** (testa)
- **Good seed** – **true to type**; healthy, **pure** (no inert matter, no weed seeds); **viable** (germination tested recently); **uniform**; **free from seed-borne disease**

### TYPES OF SEEDS

- **By cotyledons**
  - **Monocot** – **1 cotyledon** (wheat, corn, rice); germination usually **hypogeal (except onion)**; adventitious roots; **no secondary growth**; usually **endospermic**
  - **Dicot** – **2 cotyledons** (pea, bean, mango); germination usually **epigeal (except pea and gram)**; usually **non-endospermic**; secondary growth often present; flower parts in **4s or 5s**
- **By endosperm**: **albuminous** – has endosperm (castor, poppy, custard apple) | **exalbuminous** – none (gram, pea, bean)
- **By storage**
  - **Orthodox** – survive **drying and freezing**; long storage; dried to **≤10% moisture**, stored sub-freezing (Capsicum annuum, Citrus aurantifolia, grains, legumes)

    - **Recalcitrant** – **die when dried**; short storage; **chill-sensitive**; **30–70% moisture** at maturity (coffee, mango, rubber, neem, jackfruit)
- **By food**: **grains** (rice, wheat, jowar, maize, ragi) | **pulses** (grams, peas, beans) | **oilseeds** (groundnut, sesame, mustard, coconut, sunflower)

> **Note: Hypogeal** = cotyledons stay **below** ground (most monocots); **epigeal** = cotyledons come **above** (most dicots).

## SUMMARY TABLE: PARTS OF THE FLOWER

| Whorl | Position | Unit | Essential? | Notes |
|---|---|---|---|---|
| **Calyx** | 1st, **outermost** | **sepal** | non-essential | usually green; **petaloid** if brightly colored; **poly-** (free, rose) or **gamosepalous** (fused, Datura) |
| **Corolla** | 2nd | **petal** | non-essential | variously colored; **polypetalous** (rose), **gamopetalous** (Datura), **apetalous** (Brassica rapa) |
| **Androecium** | 3rd | **stamen** | **essential**, male | **anther** (pollen) + **filament** (stalk) |
| **Gynoecium** | **innermost** | **carpel** | **essential**, female | **stigma** (sticky), **style** (tube), **ovary** → **fruit**; ovary wall → fruit coat; ovules → **seeds** |

## SUMMARY TABLE: FLESHY FRUITS

| Type | Key feature | Examples |
|---|---|---|
| **Drupe** | **one seed** in a **stony endocarp**; fleshy or fibrous mesocarp; thin exocarp | cherry, peach, plum, **coconut** (fibrous) |
| **Berry** | ovary wall becomes **juicy** | grape, banana, gooseberry, tomato |
| **Hesperidium** | berry with a **leathery rind**; **septa = carpels** | orange, lemon |
| **Pepo** | berry with a **hard rind** | watermelon, squash, pumpkin, cucumber |
| **Pome** | **accessory**; core from **ovary**, pulp from **perianth**; 5+ carpels | apple, pear, mountain ash |

## SUMMARY TABLE: DRY FRUITS

| Type | Opens? | Key feature | Examples |
|---|---|---|---|
| **Legume** | dehiscent | 1 carpel; splits along **both sutures** | pea, bean (**peanut** stays shut) |
| **Follicle** | dehiscent | 1 carpel; splits along **one suture** | larkspur, columbine, milkweed |
| **Capsule** | dehiscent | **several carpels**: loculicidal (median line), **poricidal** (holes), **denticidal** (teeth), **septicidal** (septa), **silique** (long), **silicle** (short), **pyxis** (lid) | lily; poppy; corn cockle; Dutchman's pipe; mustard; plantain |
| **Achene** | indehiscent | one seed attached at **one point** | buttercup, dandelion, sunflower |
| **Nut** | indehiscent | like an achene, **very hard wall** | oak (acorn), chestnut, hazel |
| **Samara** | indehiscent | **winged**; wind dispersal | elm, maple, ash |
| **Grain (caryopsis)** | indehiscent | **seed coat fused to pericarp** | wheat, corn |
| **Schizocarp** | splits into pieces | breaks into **mericarps** | compound pistil fruits |

## SUMMARY TABLE: SEED TYPES

| Type | Key feature | Examples |
|---|---|---|
| **Monocot** | 1 cotyledon; **hypogeal** (not onion); endospermic | wheat, corn, rice |
| **Dicot** | 2 cotyledons; **epigeal** (not pea, gram); non-endospermic | pea, bean, mango |
| **Albuminous** | **has endosperm** | castor, poppy, custard apple |
| **Exalbuminous** | **no endosperm** | gram, pea, bean |
| **Orthodox** | survives drying/freezing; **≤10% moisture**; long storage | Capsicum annuum, Citrus aurantifolia, grains, legumes |
| **Recalcitrant** | **dies when dried**; 30–70% moisture; chill-sensitive | coffee, mango, rubber, neem, jackfruit |

END OF LESSON